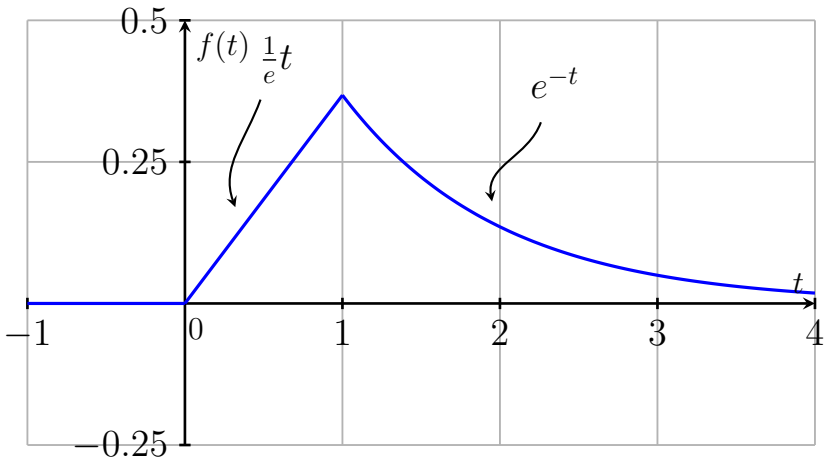
ECE 3210 hw08

1. The following figure shows the signal $f(t)$.



(a) Find the Laplace transform of this signal $F(s)$.

Solution: From the figure, the signal is $\frac{1}{e}t$ for $0 < t < 1$ and e^{-t} for $t > 1$. Use shifted unit steps to turn on each piece at the right time:

$$\begin{aligned} f(t) &= \frac{1}{e}t\,\big(u(t) - u(t-1)\big) + e^{-t}u(t-1) \\ &= \frac{1}{e}t\,u(t) - \frac{1}{e}t\,u(t-1) + e^{-t}u(t-1). \end{aligned}$$

The time-shift property needs every term written as $g(t-1)u(t-1)$, so rewrite $t\,u(t-1)$ by adding and subtracting $u(t-1)$:

$$t\,u(t-1) = t\,u(t-1) - u(t-1) + u(t-1) = (t-1)\,u(t-1) + u(t-1).$$

Substituting this, and using $e^{-t} = e^{-1}e^{-(t-1)}$ so the exponential is also a function of $t-1$:

$$\begin{aligned} f(t) &= \frac{1}{e}t\,u(t) - \frac{1}{e}\Big[(t-1)u(t-1) + u(t-1)\Big] + e^{-t}u(t-1) \\ &= \frac{1}{e}t\,u(t) - \frac{1}{e}(t-1)u(t-1) - \frac{1}{e}u(t-1) + \frac{1}{e}e^{-(t-1)}u(t-1). \end{aligned}$$

Now transform term by term using $t\,u(t) \leftrightarrow \frac{1}{s^2}$, $u(t) \leftrightarrow \frac{1}{s}$, $e^{-t}u(t) \leftrightarrow \frac{1}{s+1}$, and the time-shift property $g(t-1)u(t-1) \leftrightarrow e^{-s}G(s)$:

$$F(s) = \frac{1}{e}\left(\frac{1}{s^2} - \frac{e^{-s}}{s^2} - \frac{e^{-s}}{s} + \frac{e^{-s}}{s+1}\right).$$

(b) What is the region of convergence (ROC) for this Laplace transform?

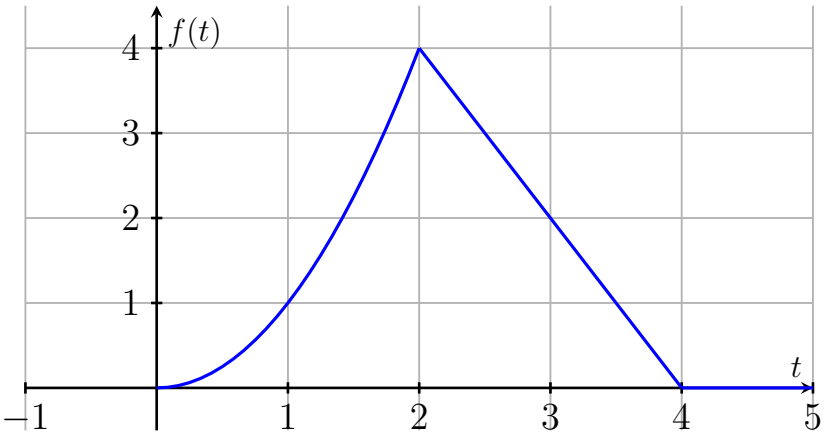
Solution: The signal does not have finite support, because the exponential e^{-t} continues for all $t > 1$, so we cannot simply say the transform converges for all s.

Check each term of $F(s)$ separately. The $e^{-t}u(t-1)$ term (giving $\frac{e^{-s}}{s+1}$) converges for $\Re(s) > -1$. All of the other terms come from signals with finite support, so they converge for all values of s.

The ROC of the sum is the intersection of the individual ROCs, so the ROC is

$$\Re(s) > -1.$$

2. The following figure shows the signal $f(t)$.



(a) Find the Laplace transform of this signal $F(s)$.

Solution: From the figure, $f(t) = t^2$ on $0 < t < 2$, $f(t) = -2t + 8 = -2(t-4)$ on $2 < t < 4$, and 0 elsewhere. Split the signal into these two pieces, $f(t) = f_1(t) + f_2(t)$, and transform each one.

First piece: $f_1(t) = t^2\big(u(t) - u(t-2)\big)$. Rather than rewriting this with shifted steps, integrate the definition directly over $0 < t < 2$:

$$F_1(s) = \int_0^2 t^2 e^{-st}\, dt.$$

Use tabular integration (differentiate t^2, integrate e^{-st}, alternate signs):

sign	derivatives of t^2	integrals of e^{-st}
+	t^2	e^{-st}
−	$2t$	$\dfrac{e^{-st}}{-s}$
+	2	$\dfrac{e^{-st}}{s^2}$
−	0	$\dfrac{e^{-st}}{-s^3}$

Multiplying along the diagonals and evaluating from 0 to 2:

$$\begin{aligned} F_1(s) &= \left[\frac{t^2 e^{-st}}{-s} - \frac{2t\, e^{-st}}{s^2} + \frac{2e^{-st}}{-s^3}\right]_0^2 \\ &= -\frac{4e^{-2s}}{s} - \frac{4e^{-2s}}{s^2} - \frac{2e^{-2s}}{s^3} + \frac{2}{s^3}. \end{aligned}$$

(Only the last term survives at the lower limit $t = 0$.)

Second piece: for this ramp we use the table instead of integrating. Write it with shifted steps:

$$f_2(t) = -2(t-4)\big(u(t-2) - u(t-4)\big) = -2(t-4)u(t-2) + 2(t-4)u(t-4).$$

The second term is already of the form $g(t-4)u(t-4)$. The first term is not of the form $g(t-2)u(t-2)$, so write $t-4 = (t-2)-2$:

$$(t-4)u(t-2) = (t-2)u(t-2) - 2u(t-2).$$

Therefore

$$f_2(t) = -2(t-2)u(t-2) + 4u(t-2) + 2(t-4)u(t-4).$$

With $t\,u(t) \leftrightarrow \frac{1}{s^2}$, $u(t) \leftrightarrow \frac{1}{s}$ and the time-shift property:

$$F_2(s) = -\frac{2e^{-2s}}{s^2} + \frac{4e^{-2s}}{s} + \frac{2e^{-4s}}{s^2}.$$

Combine: add $F_1(s) + F_2(s)$. The $\frac{e^{-2s}}{s}$ terms cancel ($-4+4=0$), and the $\frac{e^{-2s}}{s^2}$ terms combine ($-4-2=-6$):

$$F(s) = \frac{2}{s^3} - \frac{2e^{-2s}}{s^3} - \frac{6e^{-2s}}{s^2} + \frac{2e^{-4s}}{s^2}.$$

(b) What is the region of convergence (ROC) for this Laplace transform?

Solution: The signal is nonzero only on the finite interval $0 < t < 4$ and is bounded there, so the integral $\int_0^4 f(t)e^{-st}\,dt$ is finite for every s. The ROC is therefore all values of s.

3. Consider the following Lapalace transforms $F(s)$. Find the inverse Laplace transform $f(t)$ for each case.

(a)

$$F(s) = \frac{s+1}{s^2+6s+25}$$

Solution: The denominator has complex roots, so complete the square:

$$F(s) = \frac{s+1}{s^2+6s+25} = \frac{s+1}{(s+3)^2+4^2}.$$

To match the table entries for damped cosine and sine, make the numerator $s+3$ and put the leftover in a sine term with numerator 4:

$$F(s) = \frac{(s+3)-2}{(s+3)^2+4^2} = \frac{s+3}{(s+3)^2+4^2} - \frac{2}{4}\cdot\frac{4}{(s+3)^2+4^2}.$$

Using $e^{-at}\cos(\omega t)u(t) \leftrightarrow \frac{s+a}{(s+a)^2+\omega^2}$ and $e^{-at}\sin(\omega t)u(t) \leftrightarrow \frac{\omega}{(s+a)^2+\omega^2}$ with $a=3$, $\omega = 4$:

$$f(t) = e^{-3t}\left(\cos(4t) - \frac{1}{2}\sin(4t)\right)u(t).$$

(b)

$$F(s) = \frac{5}{s^2(s+1)}$$

Solution: Use partial fractions. The repeated root at $s=0$ requires both a $\frac{1}{s^2}$ and a $\frac{1}{s}$ term:

$$F(s) = \frac{5}{s^2(s+1)} = \frac{a}{s+1} + \frac{b}{s^2} + \frac{c}{s}.$$

The coefficients a and b come from the cover-up method:

$$a = (s+1)F(s)\Big|_{s=-1} = \frac{5}{s^2}\Big|_{s=-1} = 5, \qquad b = s^2F(s)\Big|_{s=0} = \frac{5}{s+1}\Big|_{s=0} = 5.$$

For c, multiply both sides by s and take $s \to \infty$. The left side goes to 0 because $F(s)$ decays like $1/s^3$:

$$\lim_{s\to\infty} sF(s) = \lim_{s\to\infty}\left[\frac{5s}{s+1} + \frac{5s}{s^2} + c\,\frac{s}{s}\right] \quad \Rightarrow \quad 0 = 5+0+c \quad \Rightarrow \quad c = -5.$$

So

$$F(s) = \frac{5}{s+1} + \frac{5}{s^2} - \frac{5}{s}.$$

Transforming each term with $e^{-t}u(t) \leftrightarrow \frac{1}{s+1}$, $t\,u(t) \leftrightarrow \frac{1}{s^2}$, $u(t) \leftrightarrow \frac{1}{s}$:

$$f(t) = 5e^{-t}u(t) + 5t\,u(t) - 5u(t) = \big(-5 + 5t + 5e^{-t}\big)u(t).$$